WORKBOOK

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Group 18

CONTENTS

Introduction
- design brief
- topic
- target group

Discover
- literature reviews
- interviews
- mobile diaries

Define
- personas
- scenarios
- POV statements
- HMW statements
- problem statement

Develop
- worst possible idea
- brainstorming
- moodboarding
- parallel design
- sketching

Deliver
- cognitive walkthrough
- cooperative evaluation

Final Design Concept
- interdependence
- problem statement
- components & interaction

Future Plans

INTRODUCTION

Design brief

"Designing for interdependent living"

Our approach

This project explores the interdependence between museum visitors and artefacts. We were inspired by the first workshop, which presented the design approach "Sustaining the ecology of care". This involves deconstructing an activity and considering all the constituents that enable it to take place. This led us to question how highlighting the life story of artefacts could change the museum experience.

We were also influenced by The Fair Tracing Project (Light, 2011), aimed at improving accountability and transparency in product supply chains. Additionally, we drew inspiration from the ongoing debate about colonialism and repatriation in museums. This motivated us to explore how ethical information about artefacts could be made more accessible to visitors in a personal and emotionally engaging way.

INTRODUCTION

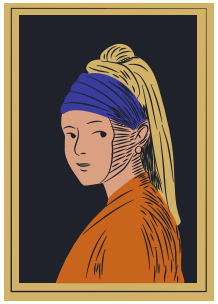
Project topic:

Human-artefact relationships &
Ethical aspects of museums

INTRODUCTION

Target group

Museum-goers

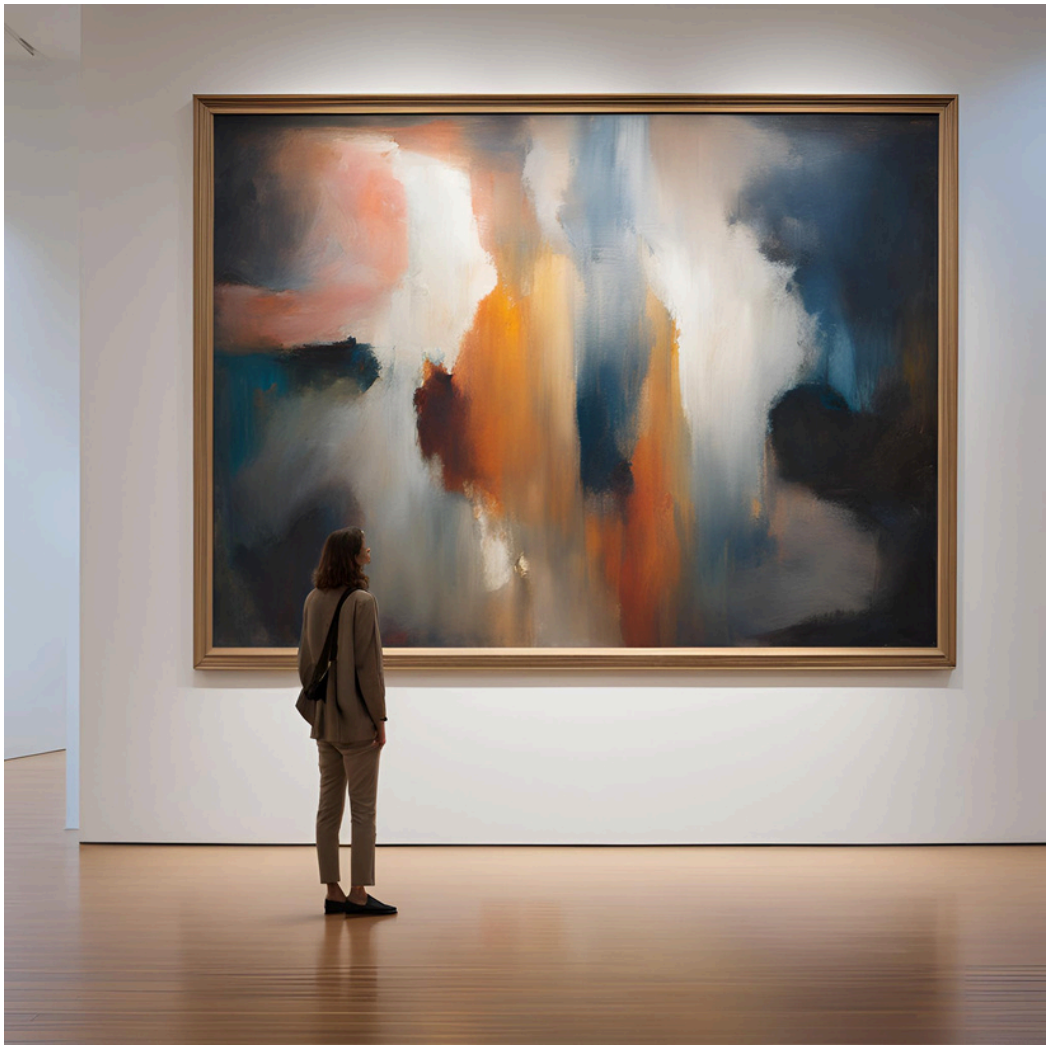


Why?

This group is easily accessible (there is a wide variety of museums in Stockholm) and we all have an interest in and prior experience of working with museums.

However...

Given more time, it could have been beneficial to also talk to museum staff, historians, curators, and researchers.



DISCOVER

Methods

- Literature review
  - To get an initial insight into the field we wanted to study
- Interviews
  - To gain firsthand and in-depth stories about previous museum visits, memories, and motivations
- Museum diaries
  - To gain an understanding of the situated and real-time museum experience

Other ideas

We considered analysing google map reviews, but decided against this due to time restraints.

LITERATURE REVIEW

Topics

- Ethical aspects of museums
- Human-artefact relationships
- Cultural user experience
- Museums in Stockholm

How did it help us?

- Taught us best practices in museum-related research
- Inspired interview topics
- Provided theoretical support for our use of museum diaries
- Gave us insight into the Stockholm museum market

LITERATURE REVIEW

Formalising and evaluating Cultural User Experience

Publisher: IEEE | Cite This | PDF

Markos Konstantakis ; Konstantinos Michalakis ; John Aliprantis ; Eirini Kalatha ; George Caridakis All Authors

Djurgården: En studie om social hållbarhet

Berglund, Tone

Södertörns högskola, Institutionen för naturvetenskap, miljö och teknik, Turismvetenskap.

2023 (Svenska)

Självständigt arbete på grundnivå (kandidatexamen), 10 poäng / 15 hp

Open Access | Article

Design of Digital Interaction for Complex Museum Collections

by Laura Loredana Micoli, Giandomenico Caruso and Gabriele Guidi *

Department of Mechanical Engineering, Politecnico di Milano, Via Giuseppe La Masa 1, 20156 Milan, Italy

* Author to whom correspondence should be addressed.

Multimodal Technol. Interact. **2020**, *4*(2), 31; https://doi.org/10.3390/mti4020031

Submission received: 10 May 2020 / Revised: 3 June 2020 / Accepted: 17 June 2020 / Published: 22 June 2020

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The Influences of Culture on User Experience

Conference paper | First Online: 19 June 2016

pp 43–52 | Cite this conference paper

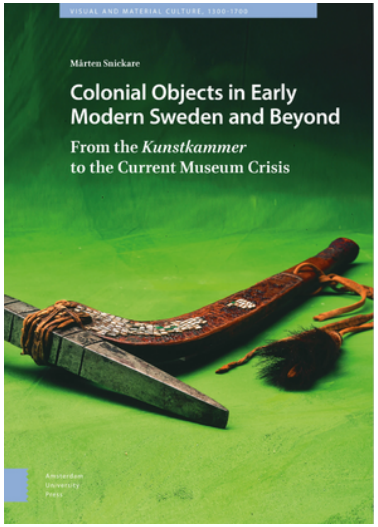
Research | Open access | Published: 28 February 2024

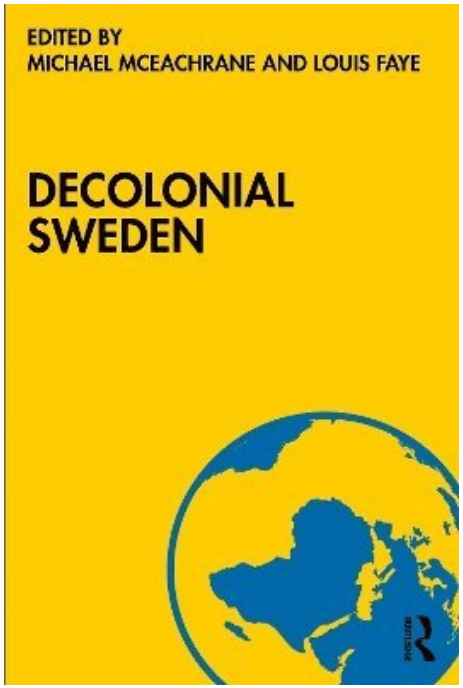
Advanced systems and technologies for the enhancement of user experience in cultural spaces: an overview

Sonia Capece, Camelia Chivăran, Giovanna Giugliano, Elena Laudante, Maria Laura Nappi & Mario Buono

Heritage Science **12**, Article number: 71 (2024) | Cite this article

2329 Accesses | **3** Citations | Metrics





RESEARCH-ARTICLE

Exploring relationships between museum artefacts through spatial interaction

Authors: Susan Ali, Ben Bedwell, Boriana Koleva | Authors Info & Claims

NordiCHI '18: Proceedings of the 10th Nordic Conference on Human-Computer Interaction • Pages 224 - 235

https://doi.org/10.1145/3240167.3240217

Published: 29 September 2018 Publication History | Check for updates

INTERVIEWS

Interview Guide: https://tinyurl.com/5pcy33py

Our approach

Semi-structured, individual, and face-to-face interviews, approximately 30 minutes.

A pilot interview was held to ensure that the interview questions were informative and relevant.

Participants

10 museum-goers within age 21-62
(age group diversity to represent most of the target audience of museums)

Topics

personal and cultural background | museum experiences | interaction with artefacts | emotional and cognitive engagement | social dynamics | the role of technology | ethical aspects

INTERVIEWS

Interview excerpt

"I've been accused of wanting to read too much and slowing down fellow museum visitors, but I like to soak myself in it somehow and learn as much as possible and feel that I've gained new perspectives."

What did we learn?

- For many, visiting museums is a social experience
- Museum-goers have different exploration styles
- Visitors do not want to be disturbed by technology (phones, apps, etc.)

MUSEUM DIARIES

What is a museum diary?

Inspired by cultural probes and diary study (Konstantakis et al., 2017).
We instructed participants to document their museum experience in real-time with text and photos or videos using their phones.

Picture from participant

Participants

5 people age 21–50 visiting museums in Djurgården (ABBA museum, Nordic museum, Fotografiska museet, Nobel Prize museum).

Insights

- The exhibit must feel genuine
- Interactive, immersive experiences are appreciated
- Entertainment is an important aspect of the museum experience
- For some participants, documenting the experience made it more memorable and enjoyable

Museum Diary Guide: https://tinyurl.com/yxztmcm3

Konstantakis, M., Michalakis, K., Aliprantis, J., Kalatha, E., & Caridakis, G. (2017). Formalising and evaluating Cultural User Experience. 2017 12th International Workshop on Semantic and Social Media Adaptation and Personalization, 1181, 90–94. https://doi.org/10.1109/smap.2017.8022675

AFFINITY DIAGRAMMING

Why?

This method encourages group consensus and collaboration while allowing for a flexible analysis of multiple types of data (text, images, videos, etc.).

Our process

Coding

- We created codes capturing the essence of sections of our data (interviews and diaries)

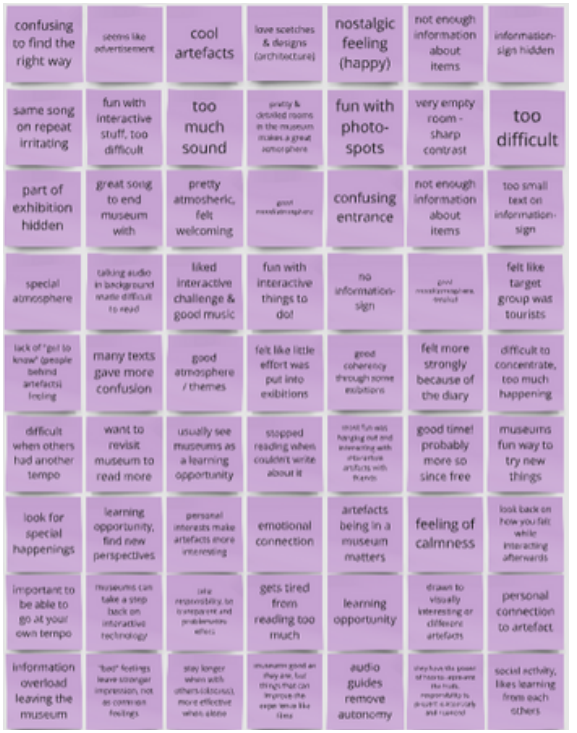
Clustering

- We individually defined low-level clusters before creating high-level clusters corresponding to general themes in a group

Ranking

- Finally, we ranked clusters to decide which key insights to focus on

Snapshot of codes



Full Affinity Diagram: https://miro.com/app/board/uXjVLfYgt8c=/?share_link_id=931084249252

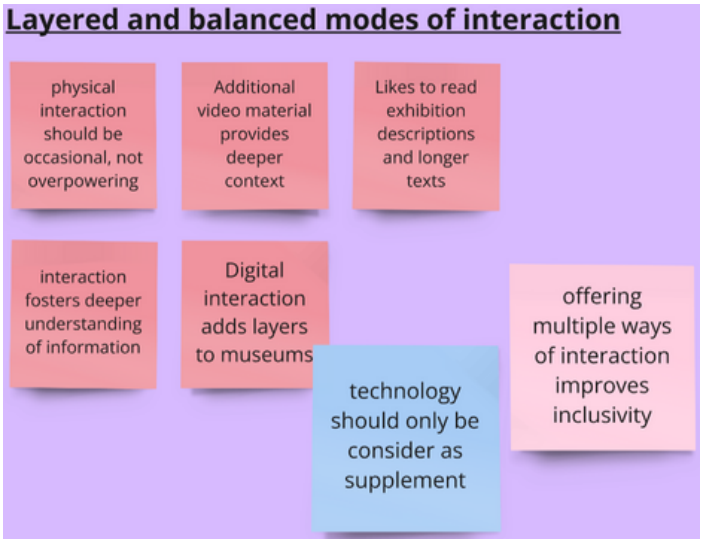
AFFINITY DIAGRAMMING

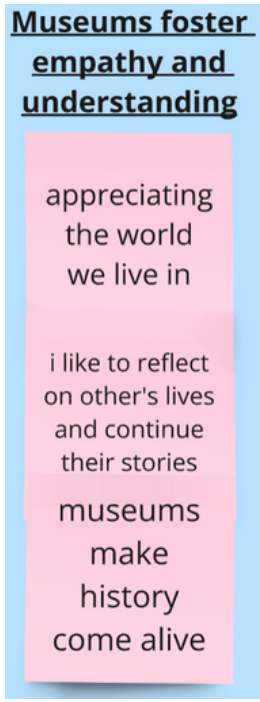
Emerging themes

Visiting museums as a social experience, the importance of humanity and personal connection, museums as a learning opportunity, parallel play in museums, etc...

Areas to dig deeper into

- Enhancing the social experience at museums
- Strengthening the emotional connection between visitors and artefacts
- Encouraging learning
- Catering to different types of visitors (educational background, expectations, exploration styles)





Snapshot of clustering

DEFINE

Methods

- Personas
  - To put a name and face for a better understanding of our users
- Scenarios
  - To contextualise the personas and identify potential pain/pleasure points in the museum experience
- POVs & How might we...?
  - To illuminate problems and opportunities
- Problem Statements
  - To define our goal

PERSONAS

Method description

To better understand how our participants might think and act, we created three personas that captured different data points: Charlie, Mary and John,

Charlie was the main persona capturing the most important data points.

Insights

- Better understanding of users
- Applied at a later stage to better evaluate our prototype
- User wants and needs
- User pain points

Full Persona Details: https://tinyurl.com/mr3kw934

PERSONAS: Charlie

Age: 23
Lives: Stockholm suburbs
Occupation: Student
Museum visits: 2-5 times/year
Traits: reflective, empathetic, curious, adventures
Current feelings: stressed, tired, busy

About:

- Busy student, struggle to find the time to visit museums
- Finds museum visits mentally stimulating
- Enjoy visiting museums with friends, but coordinating group visits is challenging
- Value the immersive experience of engaging with artifacts in person

Goals: explore culture through artifacts, engage with friends, educate themselves
Needs: break from digital overload, ways to visit museums
Pain points: not being able to learn as much as they would want to

PERSONAS: Mary

Name: Mary
Age: 37
Lives: Stockholm City Center
Occupation: Senior Consultant
Traits: studious, intuitive, empathetic, introverted

About:

- Single mother of 8 year old son
- Special interest in culture & history
- Often feels intellectually unfulfilled when visiting with others.
- Needs to visit with children due to being a mother

Goals: deepen nowledge in history, culture, art & origins of artefacts, balance information between self and children
Needs: emotional connection to artefacts
Pain points: lack of meaningful interaction, struggle to adjust to children's level

PERSONAS: John

Name: John
Age: 26
Lives: London, UK - visiting Stockholm
Occupation: Software Developer
Traits: adventurous, inquisitive, free soul

About:

- Tourist who likes to travel
- Looking for a crash course on Swedish culture in museums
- Likes sharing on social media
- Prone to information overload, unable to fully digest the information of museums

Goals: learn as much about Sweden in short time, enteratainment, share memories
Needs: concise information, visually interesting exhibits, options when learning
Pain points: information overload with text and directions, lack of diversity

SCENARIOS

Method description

For each persona, we wrote a scenario to communicate the goals of our design based on user needs, preferences and pain points. These scenarios allowed us to start considering how current technology could be used to achieve these goals.

Benefits

- Better understanding of users
- Applied at a later stage to see if scenario could be improved with our prototype

Full Scenario Details: https://tinyurl.com/3v8rah5n

SCENARIOS

Scenario 1: Charlie visits Fotografiska with their friends. Charlie is eager to explore everything and deeply reflect, but gets frustrated by being rushed by their friends, since they prefer to move quicker. Charlie ends up not being able to fully engage with the exhibit.

Scenario 2: Charlie visits Naturhistoriska alone. During this visit they connect deeply with artefacts and the visit becomes a profound exploration of identity, leaving Charlie emotionally moved and impacted.

SCENARIOS

Scenario - Mary: Mary visits the Nordic Museum with her 8-year-old son. She tries to balance her own interest while trying to also keep her son entertained. Mary struggles to explain complex issues of origin and thinks museums should offer more engaging, interactive ways to explore humanity in exhibits that are still accessible to different knowledge levels.

Scenario - John: John visits Skansen with his friends with limited time. He is overwhelmed by too many sights and excessive signage. He starts avoiding crowded areas which makes his visit less fulfilling as he misses some areas he was interested in. He wants more freedom to explore at his own pace and for the museum to be more interactive and inclusive for more people.

POVS & HOW MIGHT WE...?

Our approach

Initially, we generated as many POVs and How-Might-We-statements that we could think of. Eventually, we narrowed our options down to three POVs and HMVs to focus on 3 problem areas.

Method description

- Rephrase and contextualize the problems and opportunities found from the Personas and Scenarios
- Main points of rephrasing: either “need ... because ...” or to start with “How might we...?”

Full POVs & HMWs: https://tinyurl.com/549kr5xy

PROBLEM AREAS

Problem area 1:

Creative & appropriate way to address issues of ownership, ethical approaches and sustainability

- Museum-goers **need** appropriate ways to discuss issues of ownership and origin with those more unfamiliar with Swedish history, **because** they struggle to do so without causing emotional discomfort.

- **How might we** creatively and appropriately raise awareness about ethical issues and sustainability in museums?

PROBLEM AREAS

Problem area 2:

Cater the museum experience to different people

- Museums **need** to foster engagement among visitors of different educational backgrounds and exploration styles, **because** this would create a more inclusive experience.

- **How might we** increase the inclusivity of human-artefact relationships in museums by catering to different preferences (for instance, reading everything and slowly digesting information vs quickly moving from artefact to artefact)?

PROBLEM AREAS

Problem area 3:

Create emotional connections between artifacts and visitors from a different cultural background

- Museum-goers **need** to be able to learn about objects and exhibitions in museums in interactive and diverse ways, **because** otherwise they may experience information overload and museum fatigue.

- **How might we** foster an emotional connection between museum artefacts and people whose cultures are different to those of which the objects belong to?

PROBLEM STATEMENT

Method description

Generating the personas, scenarios, POVs and HMVs led us to our first version of problem statement:

Finding **creative** and **appropriate** ways to **address issues of ownership, ethical approaches** and **sustainability** for people of **different experience levels** and **exploration styles**, with an aim of **fostering emotional connections** and **interdependence** between **people and museum artefacts.**

Second thoughts

However, we realized it would be unrealistic to cover all the key factors of this problem statement. Hence, we narrowed it down to the second version:

Finding **creative** and **appropriate** ways to **address issues of ownership, ethical approaches** and **sustainability** in museums with an aim of **fostering emotional connections** and **personal interdependence** between **people and museum artefacts.**

PROBLEM STATEMENT

Final thoughts

Later on in the development phase, we realized there were some potential technological impracticalities and ethical issues in tracking user emotions, so we decided to solely focus on the ethical aspect of artifacts and human-artifact interdependence. Thus, the final version of the problem statement was formed:

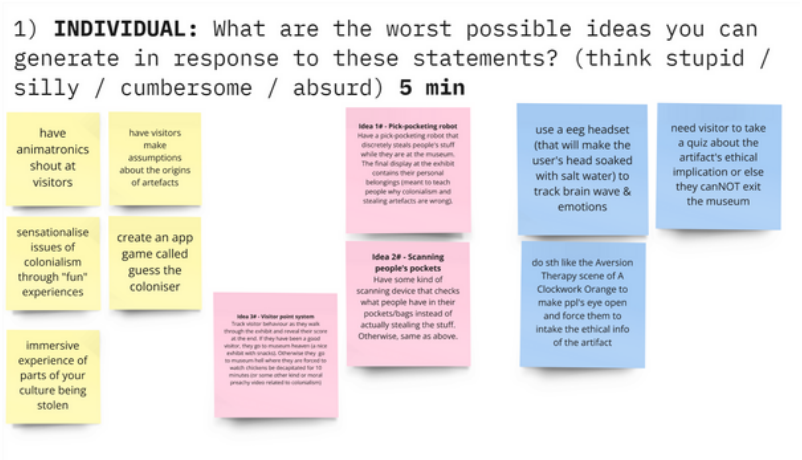
Finding **creative** and **appropriate** ways to address **ethical aspects** of exhibitions in museums with an aim of **fostering connection** and **personal interdependence** between **people and artefacts.**

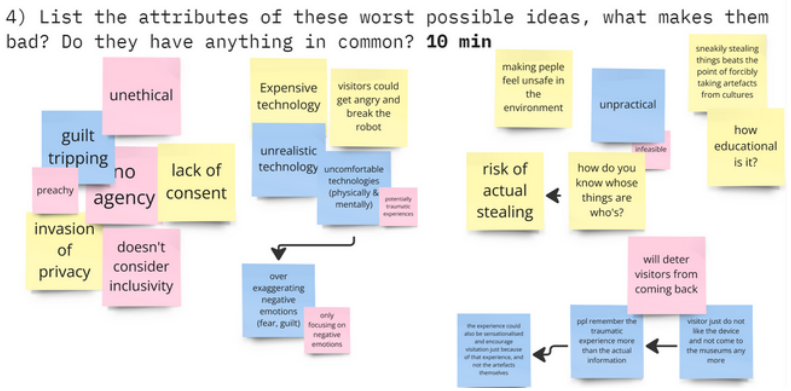
DEVELOP

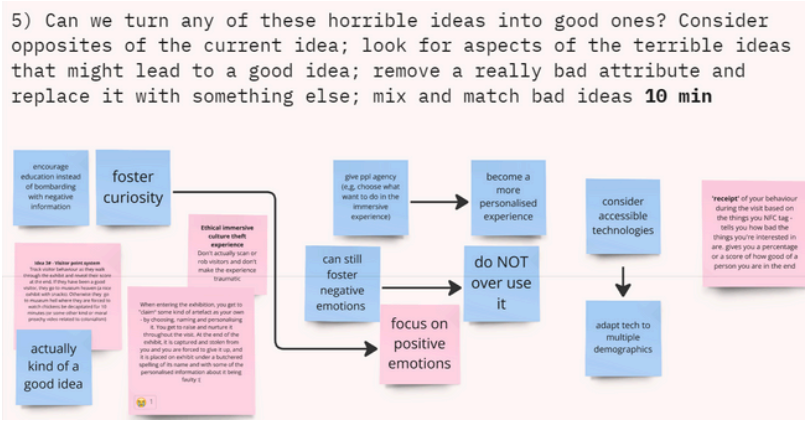
Methods

- Worst possible idea
  - To explore extreme and impractical approaches, allowing us to identify the don'ts for our design
- Brainstorming
  - To freely propose ideas and generate fresh insights to form our general design idea
- Moodboard
  - To collect visual materials to evoke our design concept
- Parallel design
  - To generate different ideas for each specific component of our design concept
- Field study
  - To better understand how visitors interact with artifacts in the museum and find inspiration for a “mini exhibit” for the prototype

WORST POSSIBLE IDEA







What to avoid in our design

infeasible, unethical, unnecessarily complex and emotionally distressing features, encroaching on user autonomy & consent

BRAINSTORMING

Rejected Idea	Reason for Rejection
Dating app to find people to go to museums with	Not aligned with the problem statement (or user needs)
EEG headset to track emotions	Salt water will be used for conduction and will soak the user's head, which may cause discomfort
Heart rate tracker to track emotions	The accuracy of using heart rate to track emotions is questionable
App that curates your museum experience to plan your visit	User's preference from data analysis shows that they do not generally like museum apps.
Gamifying the experience of learning about ethical aspects & origins of artifacts	Only focusing making the experience fun and entertaining might cause the topics to be taken less seriously
Immersive experience of historical sites with VR	The use of VR is not practical in our context and this idea already has many existing implementations

BRAINSTORMING

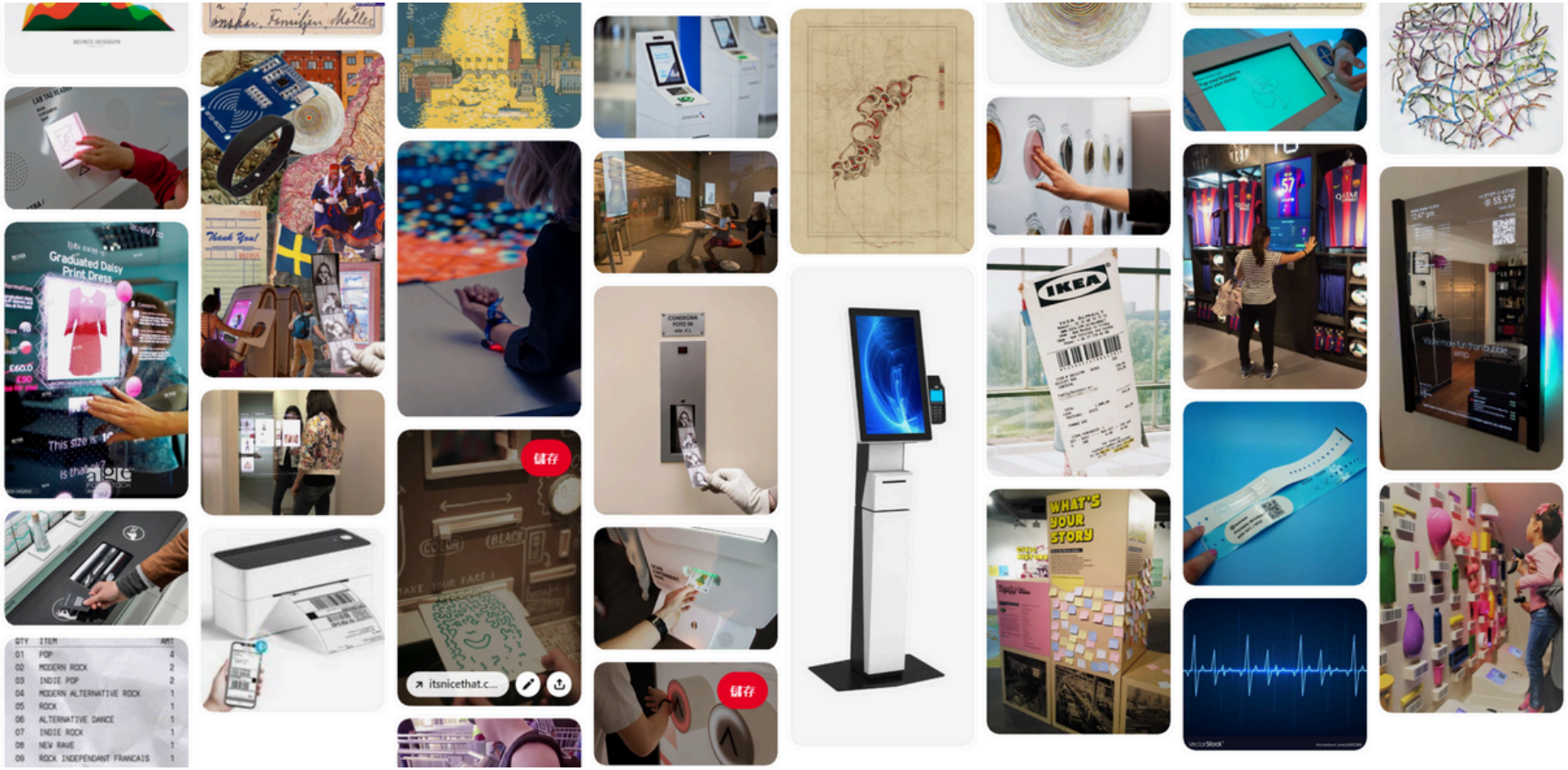
Ideas we Agree on

- Tracking visitors' routes
- A report about the visitor's museum experience
- A scanner that keeps track of the artifacts visitors resonated with the most
- Data visualization of the visit

General Design Idea

- Visitors are given a wristband with an RFID tag
- Visitors can scan artifacts they find interesting with the wristband and the data is stored in the RFID tag
- A machine at the end of the exhibition (i.e. Visit Visualizer) reads the artifact data stored in the wristband RFID tag to visualize the visitor's visit and generate a report about the artifacts

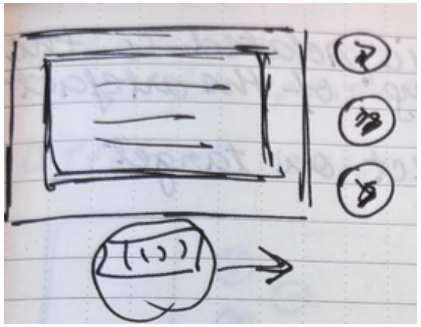
MOODBOARD



Full Moodboard in Pinterest: https://pin.it/302RERndF

PARALLEL DESIGN

Wristband & In-Exhibit Scanner



IDEATION

provide to visitors at start of experience

touch with bracelet at interesting artefacts

INTERACTIVE PROTOTYPE:

Wearable bracelet (with NFC tag)

PROS:

Non-intrusive

Feasible

CONS:

HOW TO ENCOURAGE PEOPLE TO USE IT?

+ Touching/tapping provides additional information
+ Touching/tapping saves decisions you make at interactive exhibits

(tap afterwards)

INTERACTION

+ Haptic feedback when tapping?
+ Ring that lights up around tap point?

TAP POINTS

+ Multiple tap points for the same artefact? (for different emotions?)
+ Instructions at the beginning of the museum (test tap)
+ Slider? (too complicated?)

Wristband

Materials

Paper:
- easier to wear
- fit for diff wrist size
- less hygiene issue

Silicone:
- durable
- more sustainable (?)

Data Stored

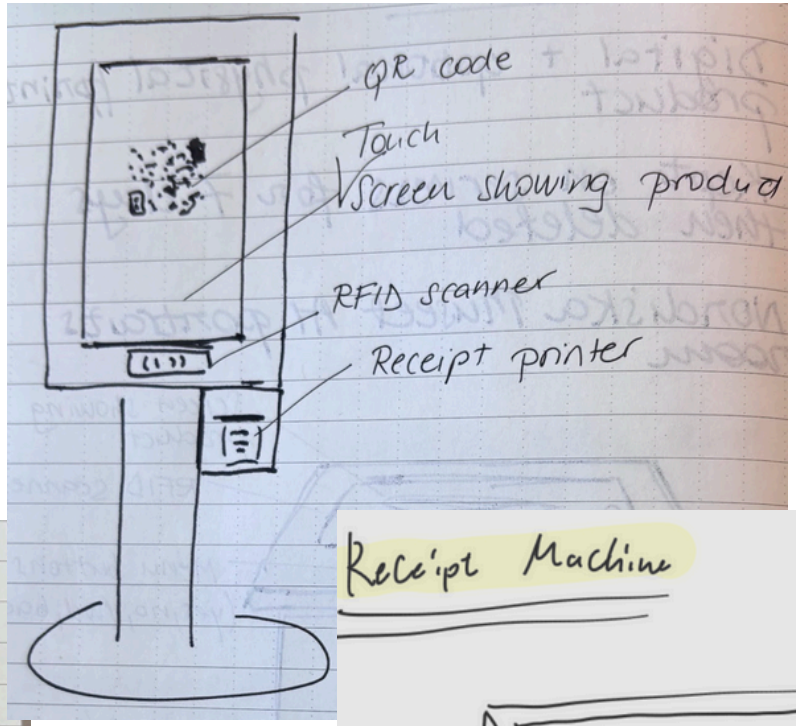
Necessary:
- Wristband ID
- which artifact interacted with
⇒ all set false & turn true when tap

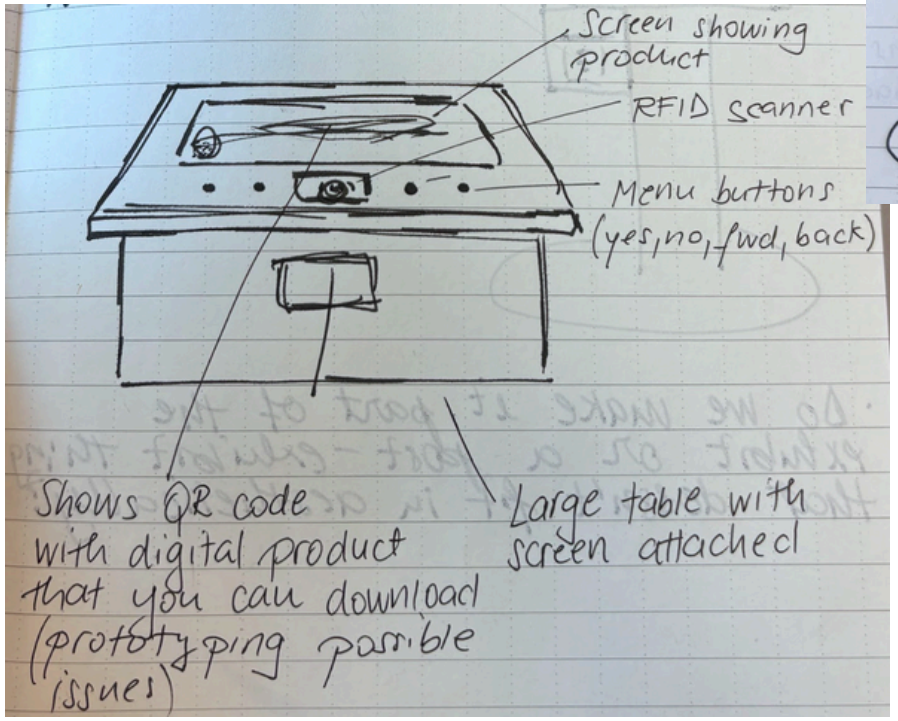
Optional:
- Order of artifact interacted
- rate at how much user like the artifact
- time of entire visit
- time stamp of tapping each artifact

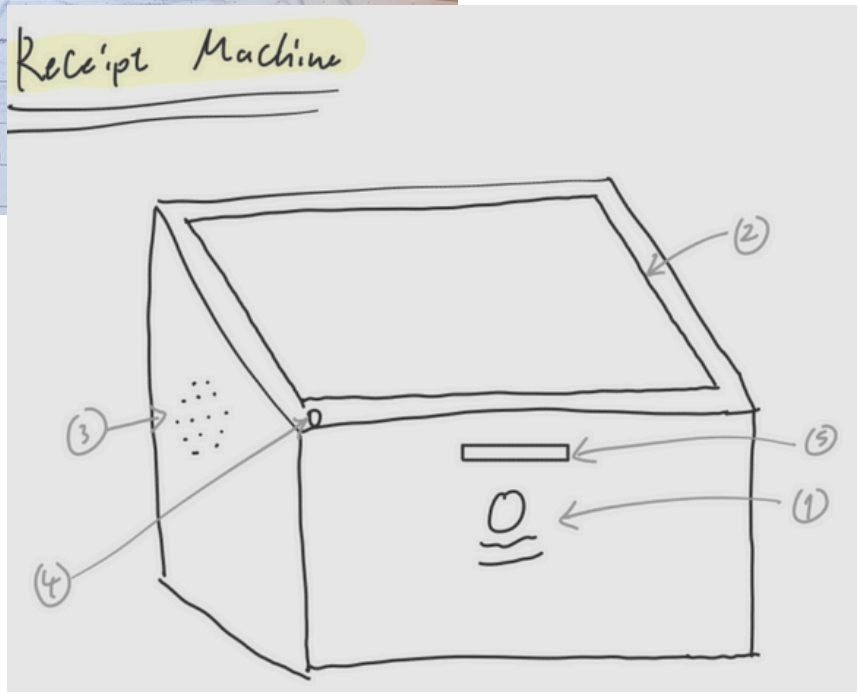
} for ranking in receipt

PARALLEL DESIGN

Visit Visualizer -- Exterior

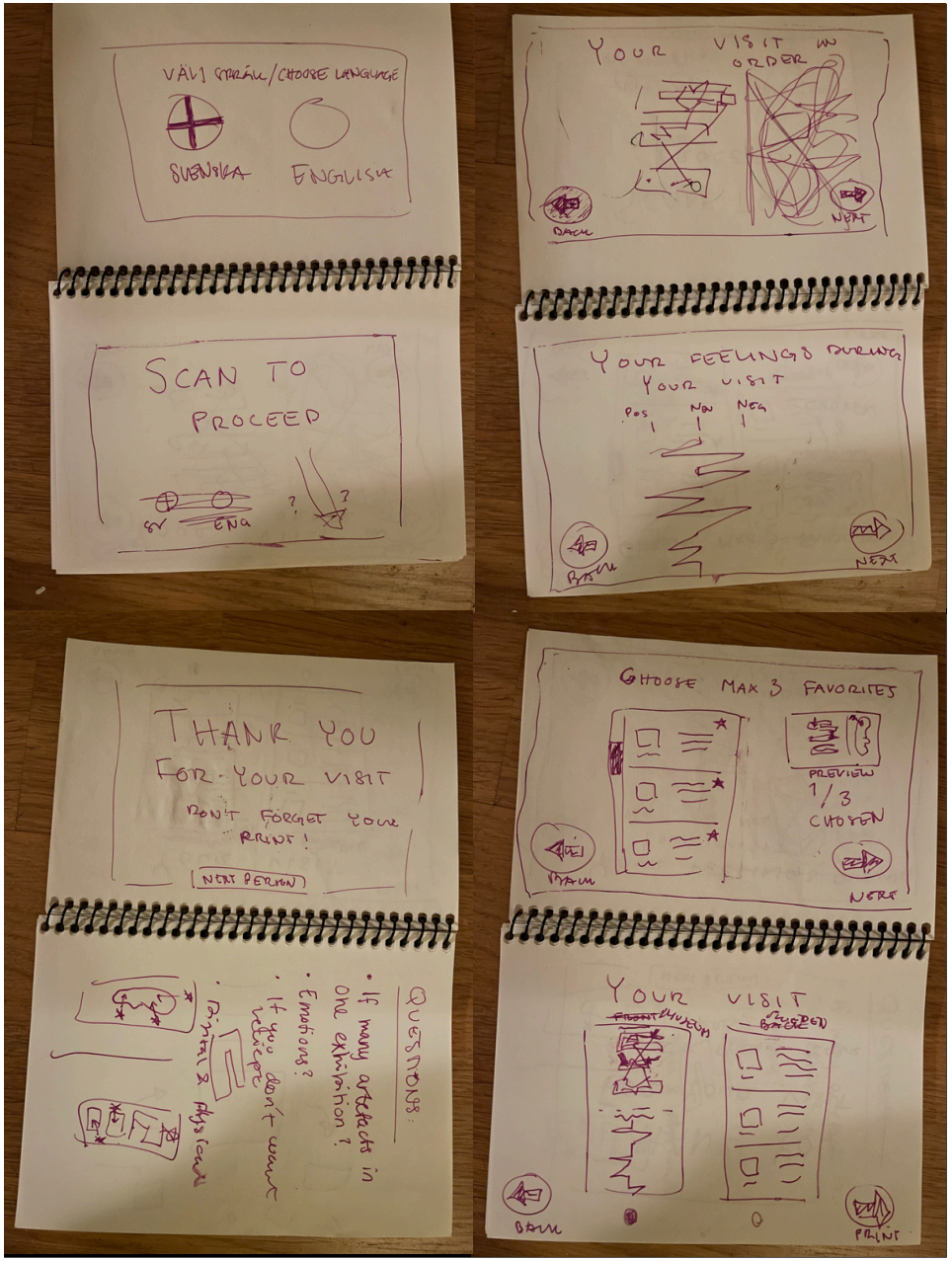






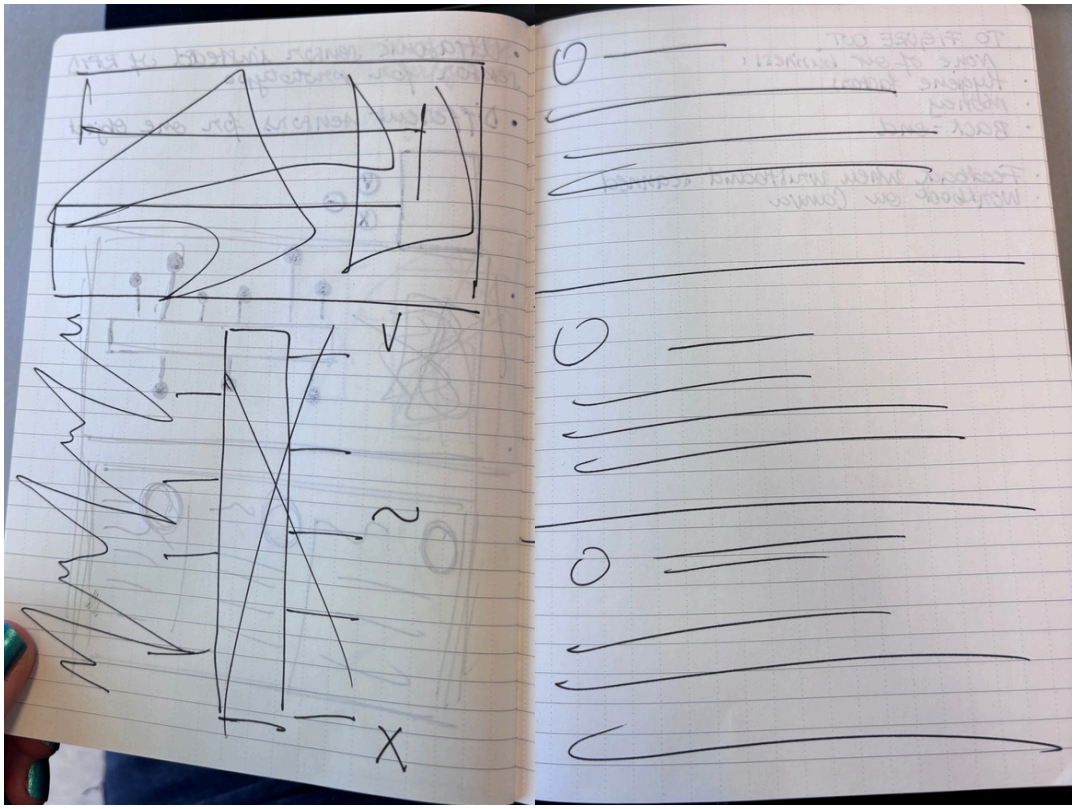
PARALLEL DESIGN

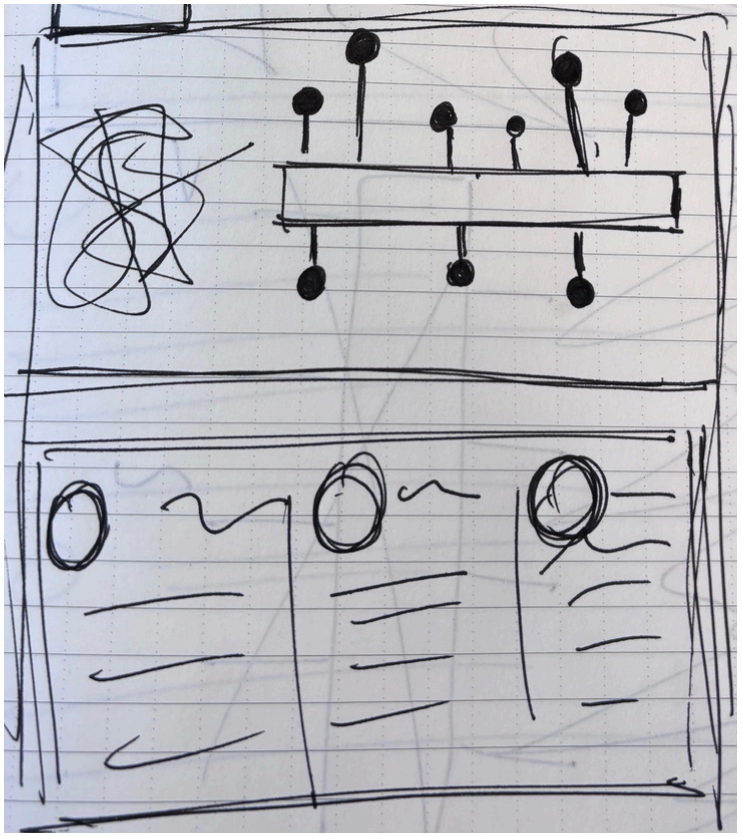
Visit Visualizer -- Interface



PARALLEL DESIGN

Visit Report

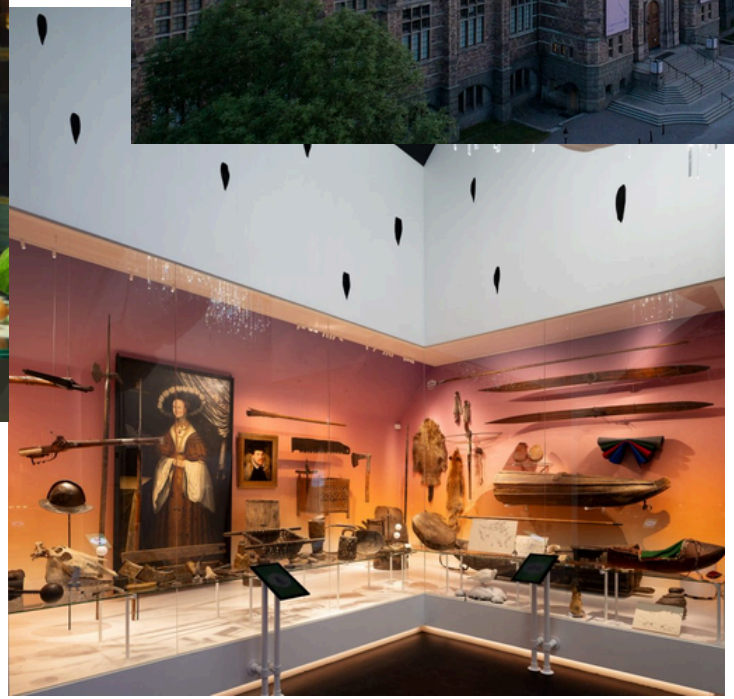




FIELD STUDY

Visit Nordiska Museet

Nordic Life exhibition

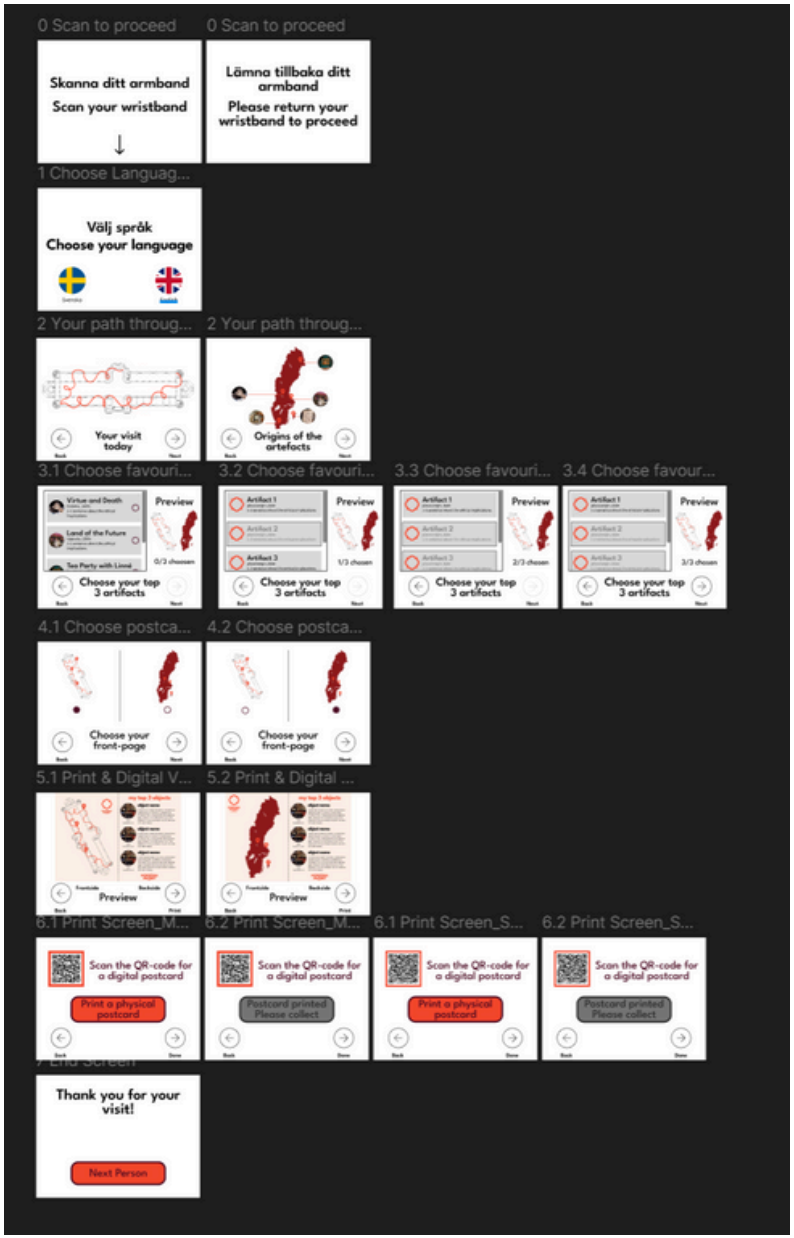


DELIVER

Methods

- Cognitive walkthrough (with designers)
  - To evaluate the entire system's experience with other designers, focusing on identifying issues and potential solutions to refine specific interactions
- Cooperative evaluation (with users)
  - To gain real-time feedback when users interact with our system to identify usability issues and gather insight about how users engage with the system

COGNITIVE WALKTHROUGH

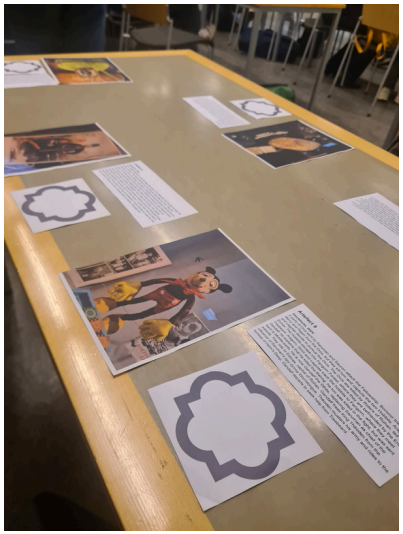


Setup

- Mid-fi Figma prototype of report machine's interface
- Paper cutout of 5 artifacts, scanner sensors, and wristbands

Task

- Explore the "museum"
- Interact with the report machine

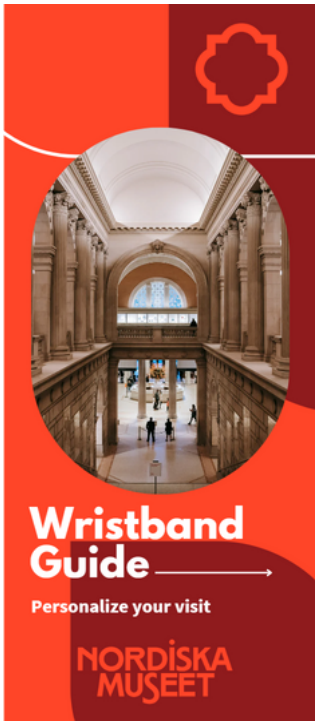


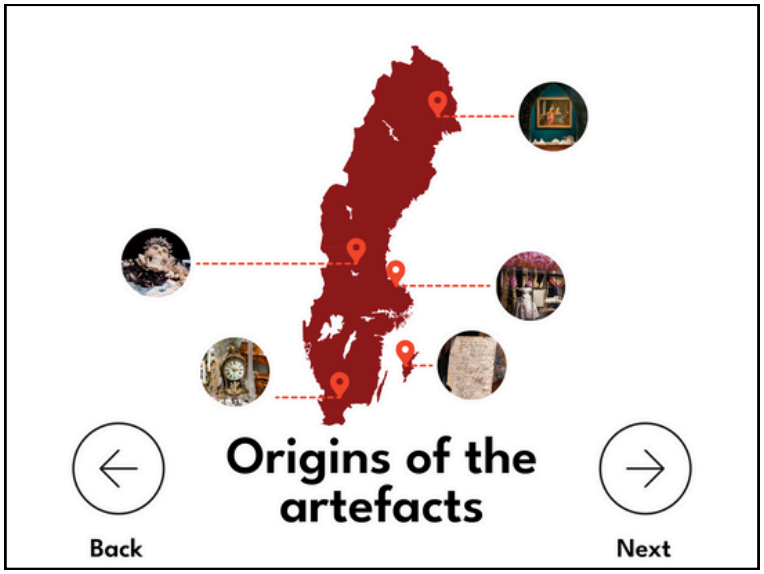
Full Figma Prototype: https://tinyurl.com/4tdf7r4m
Cognitive Walkthrough Guide: https://tinyurl.com/2jfn82ar

COGNITIVE WALKTHROUGH

Will the user try and achieve the right outcome?

- Users are interested in scanning artefacts and receiving the postcard, but not sure on why they are scanning artefacts and what they will receive at the end of their visit (3)
- Users tried to tap the pins on the maps that are suppose to be not interactive (2)





Potential Solutions

- Make an introduction pamphlet and video for the whole system
- Add pictures of the artefact on the maps that users can tap to enlarge

COGNITIVE WALKTHROUGH

Will the user notice that the correct action is available to them?

- Some users did not scan the artifacts they were interested in (2)
- User may forget to scan artefacts when they are immersed in the museum experience (1)
- Unfamiliar users may not understand that the scan icon is where they are supposed to scan their wristband (2)

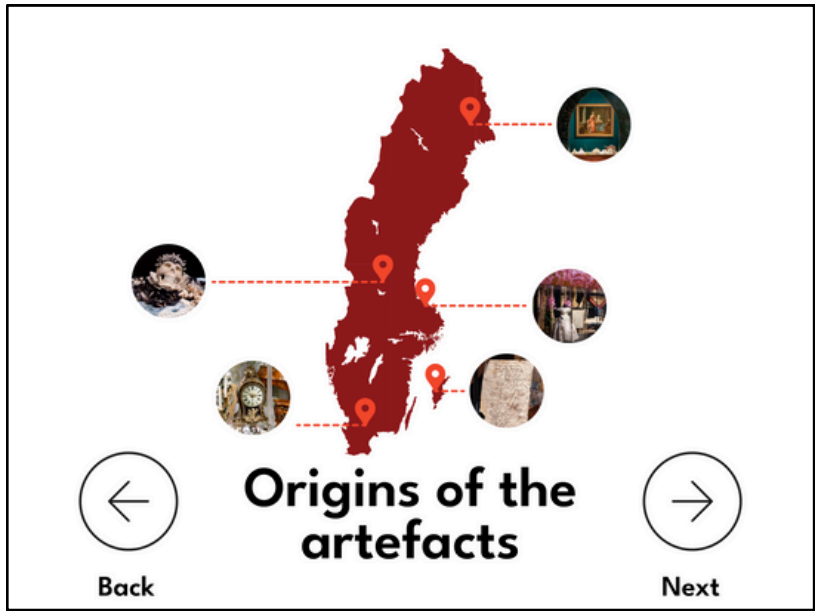
Potential Solutions

- Adding a more eye-grabbing and explanatory text/icon next to the scanner

COGNITIVE WALKTHROUGH

Will the user associate the correct action with the outcome they expect to achieve?

- Some users did not understand what the Swedish map represented (4)



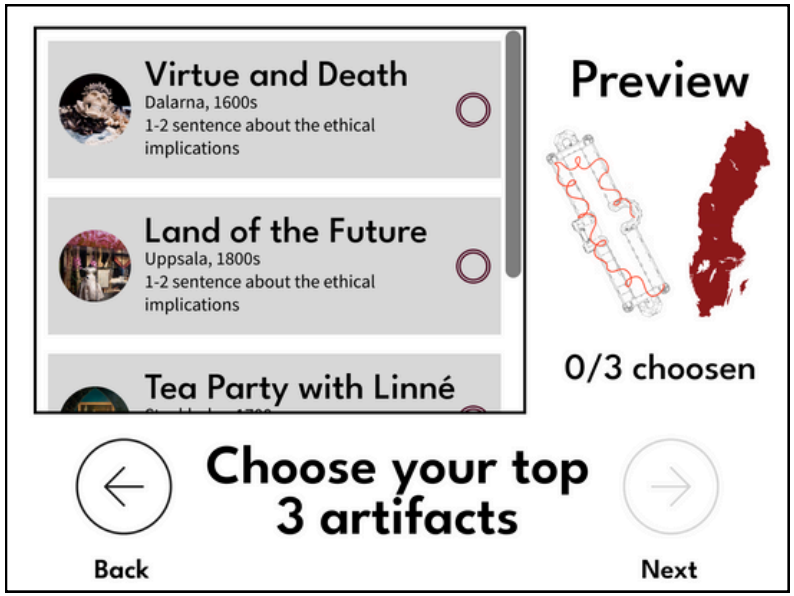
Potential Solutions

- Change the text on the screen from “Your visit today” to “Origins of the artefacts”

COGNITIVE WALKTHROUGH

If the correct action is performed, will the user see that progress is being made toward their intended outcome?

- Users did not understand that they were selecting their favorite artifacts (5)



Potential Solutions

- Add text to the screen, for instance "Choose your top 3 artefacts"

COOPERATIVE EVALUATION

Setup

- Hi-fi prototype of the Visit Visualizer interface with Arduino sensor and touchscreen display
- “Mini exhibition” environment with paper cutouts of 5 artifacts
- Introduction pamphlet
- “Fake” wristband

How did we conduct it?

- 6 participants
- Think aloud + cognitive walkthrough protocol
- Wizard of Oz part of the system (confirmation when scanning artifacts, printing, returning the wristband)

Cooperative Evaluation Guide: https://tinyurl.com/5dujzksm

COOPERATIVE EVALUATION

User Experience & Engagement

Participants had positive feedback for engagement and overall experience

- Fun and interactive
- Adds a sense of anticipation
- Personalization
- Provokes reflection

Pamphlet & Instructions

- Read by all participants
- Some participants thought the arrows on the pamphlet were confusing
  - Solution: Remove decorative arrows and keep only the ones that help indicate the page order

Scanning Process & Feedback

- Users underscored the importance of receiving feedback that the scan had been registered
  - Solution: Make the scanner light up when registering a scan

Visit Visualizer

- Intuitive UI
- Participants appreciated receiving geographical context for the artifacts' origin
- Wished for a cleaner design for the QR code & report page
- Wanted to access more details about artifacts on Sweden map (e.g. click artifact image for more information)

Report Design

- Participants generally preferred digital report to physical
- Desired more information and custimization

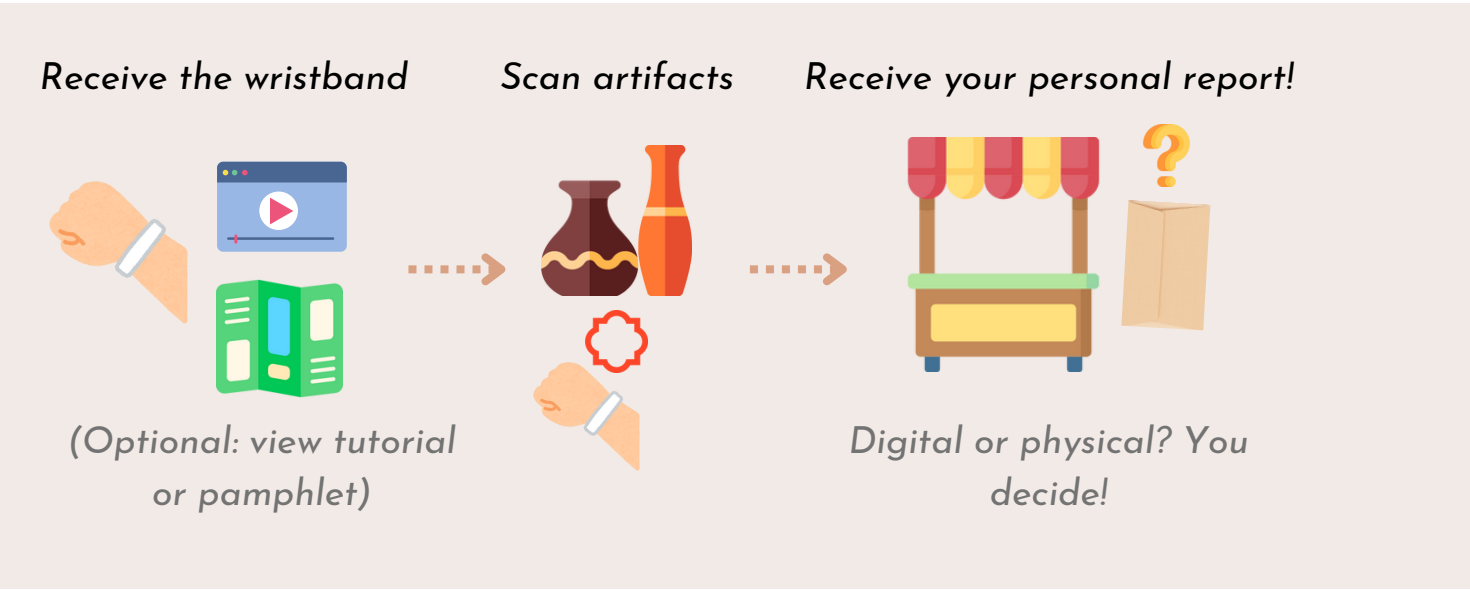
DESIGN CONCEPT

Product pitch

Personalize your visit and unlock a deeper experience with our Visit Visualizer™! While walking through the exhibit, use your wristband to scan artifacts that interest you. At the end of your visit, you will receive a personalized report based on the artifacts you selected.

Components

- Wristband
- Pamphlet
- Video tutorial
- Scanners
- Visit Visualizer
- Personal report



INTERDEPENDENCE

Creating a stronger bond between visitors and artifacts

Our product highlights the interdependence between visitors and artefacts. It unveils the journey the visitor has made through the museum parallel to the journey that artefacts have taken from their place of origin to their current location. This is intended to raise awareness of the interconnectedness of the artefacts in the exhibit. encouraging visitors to ponder their history beyond the walls of the museum.

In addition, our product facilitates a new avenue of interdependence between visitors and artefacts in the form of increasing the interactivity of the exhibit. By scanning their favourite artefacts and designing their personal report, visitors are able to curate their own experience, creating a stronger personal connection.

PROBLEM STATEMENT

*"Finding **creative** and **appropriate** ways to address **ethical aspects** of exhibitions in museums with an aim of **fostering connection** and **personal interdependence** between **people and artefacts.**"*

To appropriately and creatively address ethical aspects of exhibitions, our product makes engagement with ethical information a voluntary and explorative activity. Opting out of using the wristband does not detract from the original museum experience. The report offers a small amount of ethical information to avoid information overload, while encouraging those interested to explore further on their own.

By allowing visitors to personally scan and select their favorite artifacts, our product enhances the emotional connection between them and the artifacts. Some participants in our user research mentioned how museum details tend to fade from memory quickly, even when the visit is enjoyable. Creating a personal keepsake deepens and prolongs this connection.

WRISTBAND

Conceptual idea

- Silicone material
- Has a RFID tag that can be used to scan artifacts
- When scanning, the time and artifact ID is saved, and the scan point lights up to indicate that the interaction was successful

Prototype

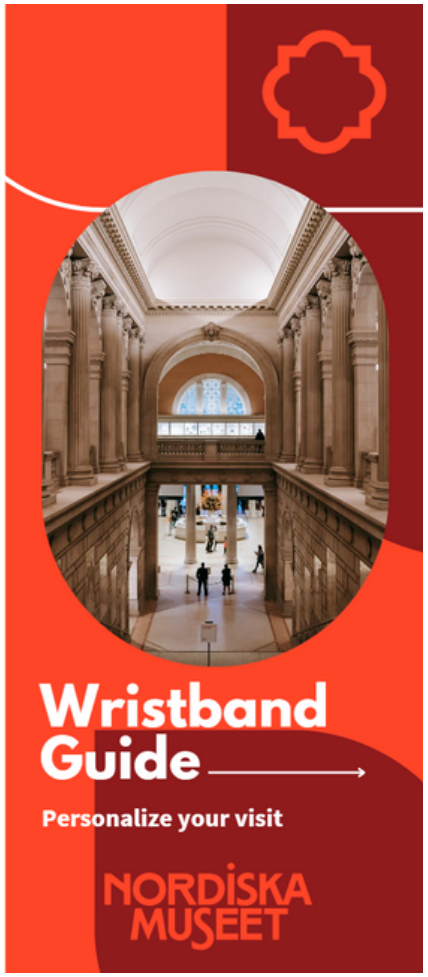
- Made out of foam band
- Does not have a built-in scanner, the interaction is registered by an ultrasonic sensor at the scan point

Idea

Execution

PAMPHLET & VIDEO TUTORIAL

Pamphlet



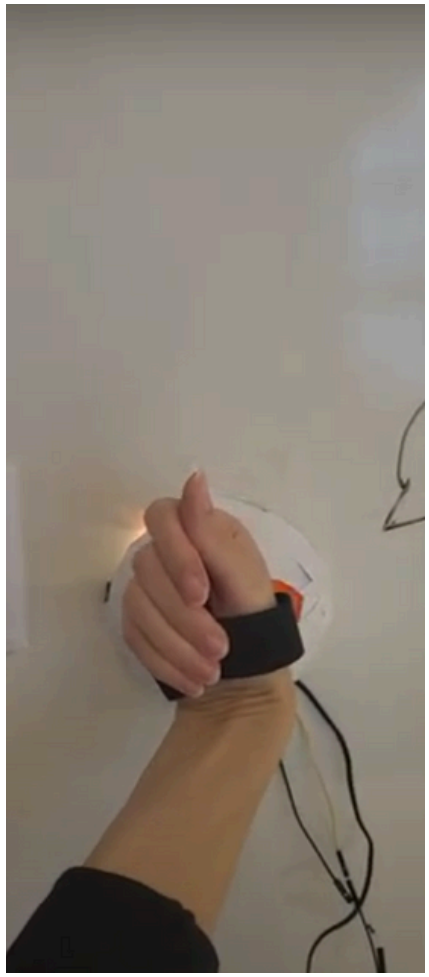
Why both?

- Different users have different preferences for receiving information
- The pamphlet contributes toward reducing congestion at the wristband station
- The video tutorial demonstrates the scanning action

Informational content

- What the wristband is used for (scanning and saving information about artifacts)
- How it is used (touching the wristband against the scanner)
- What using it results in (a personalised report)
- Where to turn if you have any problems of questions

Video tutorial



Video Tutorial:
https://tinyurl.com/yeawtajc

PAMPHLET

5 6 1

SCANNER

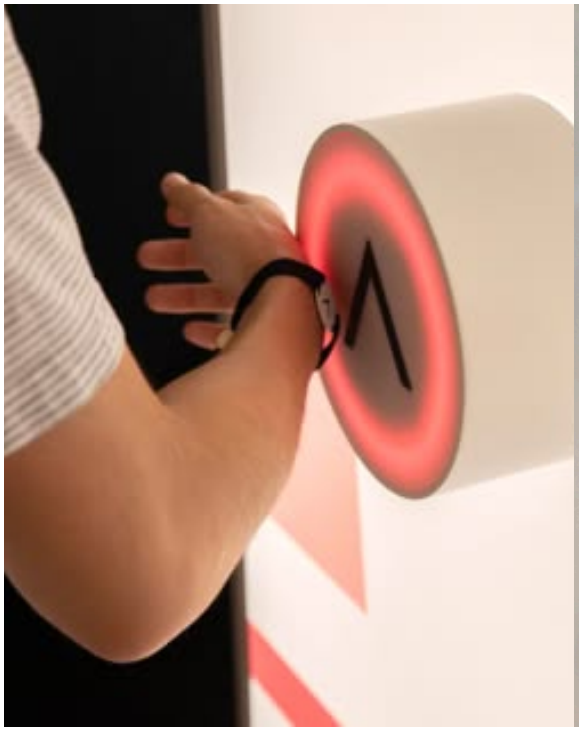
Conceptual idea

- RFID scanner that sends data to the RFID tag on the wristband
- Logo of the Nordiska Museet that lights up when the scanner is scanned

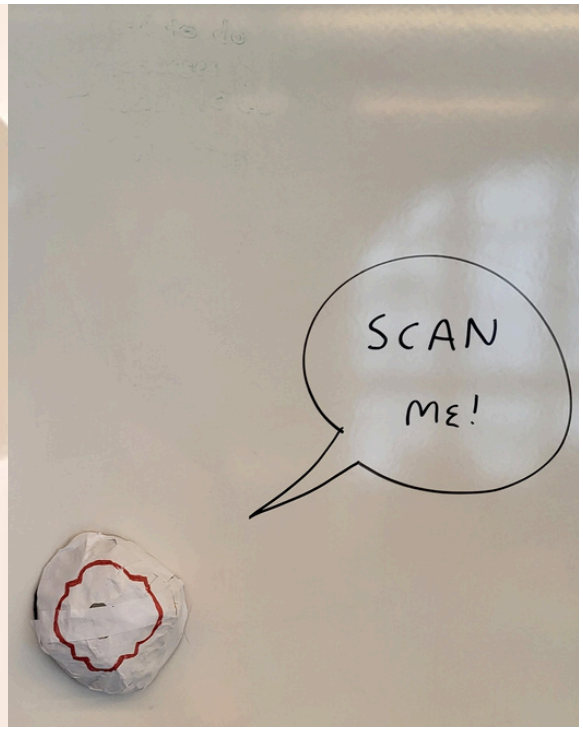
Prototype

- Paper mache dome with a cutout of the Nordiska Museet logo
- Uses ultrasound sensor and light sensor to detect the scanning interaction instead of RFID scanners
- Light flashes when scanning interaction is detected

Idea



Execution



VISIT VISUALIZER

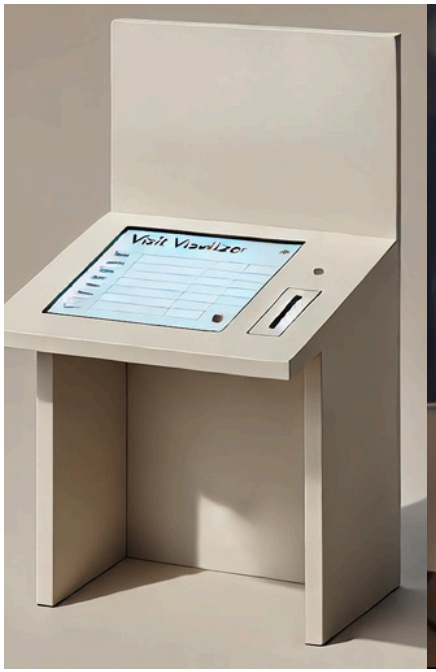
Conceptual idea

- Station where users return their wristbands and receive their personal report
- Visualizes the user's journey through the museum and from where the artifacts originate
- Users can choose a template for their report and receive it physically or digitally (or both)

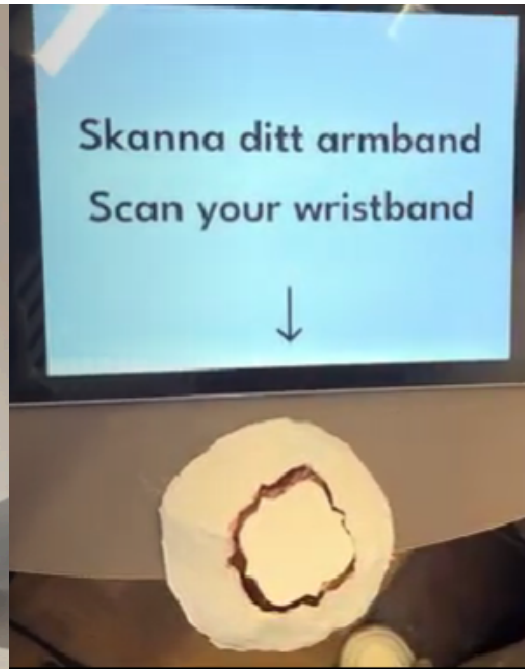
Prototype

- Built in Processing and presented on a touch-screen computer
- Lacks (automated) printing functionality

Idea



Execution



VISIT VISUALIZER INTERFACE

1

Scan your wristband

Skanna dit armband

2

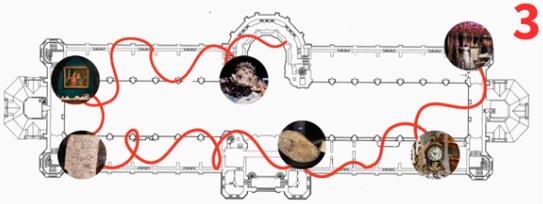
Välj språk

Choose your language

Svenska

English

3



Back

Your visit today

Next

4

Back

Origins of your artefacts

Next

5

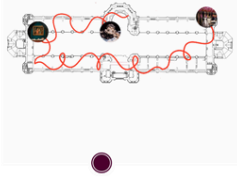
Preview
3/3 chosen

Back

Choose your favourites

Next

6



Back

Choose your front page

Next

7

Front page

Back page

Back

Preview

Next

8

Scan the QR-code for a digital report

Print a physical report

Back

Next

9

Scan the QR-code for a digital report

Report printed please collect

Back

Next

10

Thank you for your visit

Next visitor

PERSONAL REPORT

Design

- Shows either the user's path through the museum or artifact origins
- Contains additional ethical information about the user's top three artifacts
- Can be accessed through scanning a QR code or printed out as a physical souvenir

Report pages

FUTURE PLANS

If we were to develop this into a product

As this is only a prototype that was made under time restrictions, we are aware of some possible problems that could be fixed with further time to develop. These problems and ideas will be outlined in this chapter.

CURRENT CONSIDERATIONS

Accessibility

- Color blindness
- Height differences
- Less tech-savvy people
- Language

General

- Too many artefacts in museum
- Too many artefacts in one showcase
- Hygiene
- Force to hand back wristband
- Hand in wristband without getting a visualisation
- More customization & information
- Costs

Sustainability

- Digital vs printout
- Materials for wristbands

ACCESSIBILITY

Designing for accessibility

This prototype has gone through no testing for accessibility when it comes to making it more accessible for people with disabilities, and none of the designers have any disabilities. Therefore the points brought up in this part are not necessarily points that need to be fixed, but rather so discovered with the help of actual people with disabilities. There are probably more problems we have not thought about, as we don't have disabilities, but which could be discovered if we talked and worked with people with disabilities.

ACCESSIBILITY

Color blindness

The colors we have chosen for the digital part of our prototype have been directly inspired by Nordiska Museet. We have thus not looked into whether or not these would work well for people with color blindness, and would need to look into that.

Height Differences

Height differences would affect ease of use when it comes to scanning artefacts. Different mobility issues could also affect this. Thus the placement of the scanner to make it accessible to users of different heights would need to be tested.

Designing for less tech-savvy people

While developing the prototype, we did not go out of our way to include elderly or less tech-savvy participants in our data collection. During user testing we had one participant who is less familiar with technology, but more testing would be beneficial to see how the prototype could be better designed and/or explained for people less used to technology.

Language

Adding more languages to the prototype and pamphlets would make it more accessible for more people. Because of time considerations and our limited knowledge of some languges this is something to consider in the future, and not for this prototype.

GENERAL

Too many artefacts in museum

If there were a huge amount of artefacts in a museum and a person were to scan all artefacts, we are not sure how to best make the experience in the “Visit Visualizer” a pleasant one (as you need to choose only 3 from a large amount of options), or how all of this information would be stored. Thus this scenario would need to be further researched.

Too many artefacts in one showcase

If there are showcases - like in Nordiska Museet - with many artefacts in one showcase, there would have to be a smart way to efficiantly scan the right artefact without destroying the atmosphere in the exhibition. Our current idea of how to solve this is to have the scanner next to the tablets that have the information, and have the scanner scan the ID of the artefact currently shown on the scanner. Exactly how to implement this would need to be further developed and researched.

Hygiene

If the wristbands were to be reusable, the museums would (hopefully) need to clean them in some way. How this is best done is something that would need to be researched and developed. There is also a possibility to make them single-use. However, that poses a sustainability problem.

GENERAL

Force return of wristband

Ideally, the system should prevent visitors from accidentally bringing the RFID wristband home. This could be done in the scanning process for the “Visit Visualizer”, at the end of their visit. We know that this can be done, but not how we would do it within the time limit. Therefore this is a future plan.

Hand in wristband without getting a visualization

At its current state, there is no way to exit the “Visit Visualizer” without going through the complete process. In the actual product, it would be important to add a way for the user to quit at any point in the process. It has not been included due to time constraints.

More customization & information

Many of our participants expressed that they would enjoy the option to customize their visualization more. They wished to be able to choose more or fewer artifacts and get more and different information than we offered. This could be considered more if the prototype was to be developed in the future.

Costs

If the prototype was to be developed and produced, we would need to consider the costs of the project and if museums could actually implement it at that cost. However due to the goal of the project we have not yet considered this.

SUSTAINABILITY

Digital vs Physical visualizations

In our prototype we give visitors the option to chose between getting either a digital or physical visualization or both. Which of these is actually more sustainable is still up for consideration (server load vs paper). The physical option is good for the less tech-savvy people (our one less tech-savvy user tested needed instructions for how to scan the QR-code), but giving visitors more paper might just lead to them throwing it in the closest trashcan. Therefore this is still a point of discussion for the future.

Materials of the wristbands

We mainly considered two different materials for the wristbands, silicone and paper. The silicone wristbands would be reusable and possible to clean, while the paper wristbands would be single-use. We concluded that we would probably choose silicone as it would be more sustainable than throwing away lots of paper every day, but before actually implementing it we would need to research a bit more.

SWEDEN
& NORWAY
Telefon
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950855
TABLE
NO. PERSONS
WAITER
AMOUNT OF CHECK
Thank You!
YOUR PATRONAGE IS APPRECIATED
TABLE
NO. PERSONS
WAITER
CHECK NO.
950855
RFID-RC522
HYPE
PIRICA